MIL-G-17264C
19 OCTOBER 1961
SUPERSEDING
MIL-G-17264B
14 JUNE 1956

MILITARY SPECIFICATION

GENERATOR SETS, DIESEL ENGINE, STATIONARY, ALTERNATING CURRENT, 30 THROUGH 500 KW (FOR FLOATING DRYDOCKS)

This specification has been approved by the Department of Defense and is mandatory for use by the Departments of the Army, the Navy, and the Air Force.

1. SCOPE

1.1 Scope. This specification covers 450 volt, 3 phase, 60 cycle, 0.8 power factor, alternating current (ac), diesel-engine-driven generator sets for use as main and auxiliary power sources on floating drydocks where operation under the conditions of permanent list, permanent trim in the longitudinal axis, and momentary rolling are encountered. Electrical insulation protection and corrosion resistance, when exposed to a tropical salt and moisture laden atmosphere, shall be inherent within the generator set.

1.2 Classification. The generator set shall be of the 30-, 60-, 100-, 150-, 200-, 250-, 300-, 400-, or 500-kilowatt (kw) size, as specified.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

F–F–351	— Filters and Filter Elements; Lubricating Oil; Internal Combustion Engines.
FF–B–171	— Bearings, Ball, Annular (General Purpose).
TT–E–489	— Enamel, Alkyd, Gloss (for Exterior and Interior Surfaces).
VV–F–800	— Fuel Oil, Diesel.

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MIL–V–173	— Varnish, Moisture- and Fungus-Resistant (for the Treatment of Communications, Electronic, and Associated Electrical Equipment).

FSC 6115

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MIL-T-656 — Thermometer, Self-Indicating, Liquid-in-Glass (Navy Type).

MIL-T-704 — Treatment and Painting of Materiel.

JAN-P-735 — Primer, Paint, Zinc-Chromate, Alkyd Type.

MIL-E-917 — Electric Power Equipment, Basic Requirements for (Naval Shipboard use).

MIL-V-1137 — Varnish, Electrical-Insulating (for Electro - Motive Equipment).

MIL-C-2194 — Cables, Power, Electrical, Reduced Diameter Type, Naval Shipboard.

MIL-R-2729 — Regulator Sets, Voltage, A.C. Generator, Naval Shipboard use.

MIL-F-3623 — Filter Elements, Lubricating Oil, Engine (Large).

MIL-B-3743 — Brushes, Electrical Contact; and Carbon Stock, Electrical Contact Brush.

MIL-L-9000 — Lubricating Oil, Internal Combustion Engine, Diesel.

MIL-I-11683 — Interference Suppression, Radio, Requirements for Engine Generators and Miscellaneous Engines.

MIL-M-15071 — Manual, Technical, for Mechanical and Electrical Equipment (Less Electronics).

MIL-B-15072 — Batteries, Storage, Lead-Acid, Portable (Except for Aircraft and Automotive Vehicles).

MIL-E-15090 — Enamel, Equipment, Light-Gray (Formula No. 11).

MIL-R-15109 — Resistors and Rheostats, Shockproof (Naval Shipboard use).

MIL-M-15337 — Mufflers, Exhaust, Internal Combustion Engine.

MIL-C-15730 — Coolers, Fluid, Industrial, Naval Shipboard: Lubricating Oil, Hyraulic Oil, and Fresh Water.

MIL-S-16032 — Switches: Pressure, Thermostatic, and Liquid Level-Shipboard Alarm System.

MIL-P-16304 — Pyrometers, Indicating.

MIL-I-16910 — Interference Measurement, Radio, Methods and Limits; 14 Kilocycles to 1,000 Megacycles.

MIL-C-19420 — Charger, Battery, Metallic Rectifier, Naval Shipboard General Service.

MIL-F-20627 — Filters, Diesel-Fuel-Oil (Navy Standard).

STANDARDS

MILITARY

MIL-STD-129 — Marking for Shipment and Storage.

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MIL-STD-130 — Identification Marking of U. S. Military Property.

MS 35758 — Cups, Grease, Manual Feed.

MS 35759 — Cups, Grease, Automatic Feed.

DRAWING

BUREAU OF SHIPS

B-204 — Diesel Engine Fuel and Lubricating Oil Filters and Strainers Diagrammatic Piping Arrangement.

The following additional documents are required only when level A preservation and packaging, and level A or B packing are required.

SPECIFICATIONS

FEDERAL

TT-P-664 — Primer, Coating, Synthetic, Rust-Inhibiting, Lacquer-Resisting.

UU-T-81 — Tags, Shipping and Stock.

PPP-B-601 — Boxes, Wood, Cleated-Plywood.

PPP-B-621 — Boxes, Wood, Nailed and Lock-Corner.

PPP-T-60 — Tape; Pressure Sensitive Adhesive, Waterproof—for Packaging and Sealing.

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MIL-C-104 — Crates, Wood; Lumber and Plywood Sheathed, Nailed and Bolted.

MIL-P-116 — Preservation, Methods of.

MIL-B-121 — Barrier Material, Grease Proofed, Waterproofed, Flexible.

MIL-R-196 — Repair Parts for Internal Combustion Engines, Packaging of.

MIL-B-233 — Boxes, Repair Parts.

MIL-C-3600 — Compressors, Air and Gas (Except Oxygen), Packaging of.

MIL-E-10062 — Engines, Spare or Installed (Other Than Aircraft), Preparation for Shipment and Storage of.

MIL-P-14105 — Paint, Heat-Resisting, Olive Drab (for Steel Surfaces).

MIL-H-15424 — Hand Tools; Packaging of.

MIL-P-16298 — Preservation, Packaging, Packing, and Marking of Electric Machines Having Rotating Parts and Associated Repair Parts.

MIL-E-17555 — Electronic and Electrical Equipment and Associated Repair Parts, Preparation for Delivery of.

(Copies of specifications, standards, and drawing required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

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NATIONAL BUREAU OF STANDARDS

Handbook H28 — Screw-Thread Standards for Federal Services.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington 25, D. C.)

AMERICAN BUREAU OF SHIPPING (ABS)

Rules for Building and Classing Steel Vessels. (1959)

(Copies of ABS publications may be obtained from the American Bureau of Shipping, 45 Broad Street, New York 4, New York.)

AMERICAN INSTITUTE OF ELECTRICAL ENGINEERS (AIEE)

No. 45 — Recommended Practice for Electric Installation on Shipboard. (1958)

No. 606 — Recommended Specification for Speed Governing of Internal Combustion Engine-Generator Units. (1959)

(Copies of AIEE publications may be obtained from the American Institute of Electrical Engineers, 33 West 39th Street, New York 18, New York.)

DIESEL ENGINE MANUFACTURERS ASSOCIATION (DEMA)

Standard Practices for Stationary Diesel and Gas Engines. (1958)

(Copies of DEMA publications may be obtained from the Diesel Engine Manufacturers Association, 2000 K Street, N. W., Washington 6, D. C.)

3. REQUIREMENTS

3.1 Description. The generator set shall consist of an ac generator, with exciter and voltage regulation system, driven by a diesel engine. The engine shall be complete with all the accessories and auxiliaries necessary for normal operation. The set shall be suitable for installation on a floating drydock and be designed for operation with one or more similar sets in parallel. General characteristics shall be as shown in table I.

TABLE I. *General Characteristics of Generator Sets for Floating Drydocks*

Gen. size (kw)	Continuous output		General overall size**				Max. fuel consumption (lbs./hp. hr.)	Type of cranking system***
	Max. (rpm)	Max bmep* (psi)	Lg. (in.)	W (in.)	Hgt. (in.)	Wt. (lbs.)		
30	1200	135	115	40	80	9000	0.50	E A H
60	1200	135	120	46	90	9000	.50	E A H
100	1200	135	140	54	100	15000	.50	E A H
150	1200	135	155	60	120	18000	.50	A H
200	1200	135	175	60	120	22000	.42	A H
250	900	135	200	70	120	32000	.42	A H
300	900	135	210	70	126	35000	.42	A H
400	720	135	234	76	140	45000	.42	A H
500	720	135	250	85	160	50000	.42	A H

Minimum number of cylinders shall be three.
*Maximum permissible brake mean effective pressure (bmep) for naturally aspirated engines shall be 85 pounds per square inch (psi).
**These requirements are intended to be maximum. "Hgt" is the height above the deck to remove piston, cylinder head, or liner whichever is greatest.
***E — indicates electric; A — indicates compressed air; H — indicates hydraulic.
Weight does not include large unattached accessories such as coolers, starting equipment, filters, etc.

3.2 Material. Material shall be as specified herein. Material not definitely specified shall be of the quality regularly used by the manufacturer in the product he sells on the commercial market. Material shall be free from all defects and imperfections that might affect the serviceability and appearance of the finished product. The engine bedplate, cylinder block, and crankcase shall be of welded or cast steel, cast iron with a minimum tensile

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strength of 25,000 psi or, for generator sets smaller than 100 kw, may be of high tensile aluminum alloy. Portions of the main bearing supports and caps which are not part of the engine base shall be of forged or cast steel. Where used, aluminum alloy parts shall be given a protective treatment by an anodic oxidation process or other suitable protection against corrosion. In general, generator metal parts shall be of steel, malleable iron, nodular iron, brass, or bronze. Aluminum will be permitted for handhole covers and for lubricant seals for sleeve bearings. Where steel is used for small parts or fastenings such as bolts, nuts, springs, or brushholder studs, it shall be satisfactorily treated for corrosion resistance. Insulation shall be class B for all generators and exciters, impregnated and treated in conformance to Specification MIL-E-917.

3.3 **Identical items.** All units of the same classification furnished under any specific contract shall be physically and mechanically identical. This requirement includes parts, assemblies, components, and accessories. No deviations will be acceptable without prior written approval.

3.4 **Preproduction model.** As soon as practicable after award of the contract or order and prior to submission of any complete generator sets for final acceptance, the contractor shall furnish a preproduction model for inspection to determine conformance to this specification. Approval of the preproduction model by the activity concerned shall not relieve the contractor of his obligation to supply generator sets conforming to this specification. No changes or deviations from the preproduction model will be acceptable without prior written approval from the contracting officer. The preproduction model may be the first complete generator set offered.

3.5 **Safety.** All equipment or exposed items of the generator set which are energized electrically, and all rotating or reciprocating parts which are of such nature or so located as to become a hazard to operating personnel shall be insulated, fully enclosed, or properly guarded.

3.6 **Operating conditions.** Generator sets shall be designed for continuous operation under any combination of the following conditions:

(a) At any load up to full rated load.

(b) A 15 degree permanent list, 5 degree permanent trim in the longitudinal axis of the set, and momentary rolling of 35 degrees.

(c) In a moisture and salt laden atmosphere.

(d) Within the temperature range of 32 degrees Fahrenheit (F) to 125 degrees F.

3.7 **Design.** The design of the generator set shall be in accordance with the best engineering practice. The generator set design and accessory installations shall permit accessibility for maintenance and service in the field. Replacement and adjustment of component parts and accessories shall be possible with minimum disturbance to other elements. Drainage shall be in accordance with the best engineering practice. All screw threads shall conform to Handbook H28. The generator rotor shall be directly connected and in line with the engine crankshaft. Gear, belt, or chain drives will not be acceptable. Except for accessories specified for mounting elsewhere, the complete generator set shall be assembled as a unit on a welded steel subbase, or with the generator on an extension of the engine bed or crackcase. The subbase shall be designed not only for assembling, handling, and shipping the generator set as a unit, but also to provide the strength and stiffness needed to maintain alinement of the set during rolling and pitching of the dock. The dynamic forces which must be resisted shall be equal to at least four times the value obtained by use of the formulae specified below. The necessary generator set data required to calculate the stress on the bearing

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members and supporting foundations in the dock, under full dead load combined with dynamic action induced by rolling in a mountainous seaway, in accordance with the formulae specified below, shall be furnished by the generator set manufacturer. The complete generator set shall withstand the dynamic forces induced under the above conditions when computed in accordance with the following formulæ:

$$F = \frac{Wl}{32.2}\left(\frac{4\pi^2\theta}{180T^2}\right)$$

$$T = \frac{1.108K}{\sqrt{GM}}$$

Where F = dynamic force.

T = the period of complete roll of the dock from one side to the opposite and back again in seconds (to be furnished by the Government).

W = the weight of the unit in pounds.

l = the distance in feet between the center of rotation of the dock and the center of gravity of the generator set (to be furnished by the Government).

K = the mass radius of gyration of the dock in feet (to be furnished by the Government).

θ = the angle of roll in degrees from the upright position (see 3.6 (b)).

GM = the metacentric height of the dock in feet (to be furnished by the Government).

3.7.1 *Coupling.* Major coupling parts shall be of cast or forged steel. The method of connecting the engine and generator shafts together shall transmit the torque smoothly and compensate for any misalinement between shafts without causing injurious stresses and loads on the generator or the engine. Couplings have backlash requiring lubrication or containing parts adversely affected by military fuels or lubricants, or maximum operating temperatures are prohibited. Flexible couplings for two-bearing generators shall compensate for end, angular, and lateral movement of the coupled shafts without either shaft exerting a thrust on the other and allowing each to rotate in its normal axial position.

3.7.2 *Performance requirements.* Each generator set shall be capable of delivering its specified rated kw continuously, and 110 percent rated kw for 2 hours within any 24 hour period, without injury to any part. The term continuous duty as used herein is defined as a requirement of service demanding operation at a substantially constant load for an indefinitely long period. The rated revolutions per minute (rpm) of the generator set shall conform to the restrictions of table I and the requirements of the generator for 60 cycle operation. The mass elastic system of the engine together with all other connected masses rotating with it, shall have no dangerous torsional vibration within the speed range of 10 percent below rated rpm to the setting of the overspeed trip device. The permissible vibratory stress, S_v, shall be no greater than

$$S_v = \frac{\text{Torsional limit of shaft material}}{3k_t}$$

where k_t is the maximum torsional stress concentration factor (ratio of the maximum stress to the nominal stress) caused by shaft fillets, keyways, oil holes, surface finish, or other abrupt change in shaft cross section. For critical speeds below the speed range defined above, the superimposed vibratory stresses shall not exceed twice the permissible S_v defined above. Generators, when driven by their intended prime movers, shall operate in parallel in accordance with the following:

(a) At any given load division, the actual kw load of any generator, expressed as a percentage of its continuous kw rating, shall not differ from the total system kw load, expressed as a percentage of the total continuous kw rating

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of all connected generators, by more than 5 percent for generators of the same size, design, and rating or by more than 10 percent for generators of unequal size, design, and rating.

(b) The maximum difference in reactive kilovolt-amperes (kvar) supplied by one generator from that supplied by another generator shall not exceed 10 percent of the continuous kilovolt-amperes rating of either generator at any load from zero to 100 percent of the continuous kva rating of the two generators operating at rated frequency and at any power factor between zero lagging and unity power factor. The circulating current between generators operating at no load shall not exceed 6 percent of the continuous current rating of either generator at any frequency from plus or minus 5 percent of rated frequency.

3.8 **Engines.** Engines shall be of the multicylinder, liquid-cooled, 2- or 4-stroke cycle, full diesel type, and shall be suitable for continuous operation at the generator rated full load speed. Engines shall be furnished complete with all accessories required for proper operation. The rated continuous horsepower output of the engine shall be sufficient to drive its accessories and the generator at 100 percent load. Limiting values for bmep, and rpm, together with approximate dimensions, shall be according to table I. Engine exhaust shall be clear, in the sense normally accepted commercially when the engine is operating continuously at 105 percent of its rated brake horsepower (bhp) output and at rated rpm, using fuel oil conforming to Specification VV–F–800. The engine shall be equipped with removable cylinder liners and shall be designed to operate with closed fresh water cooling system.

3.8.1 *Accessories.* Accessories shall include engine-driven, centrifugal, fresh water pump; engine-driven, centrifugal, sea water pump, when specified; lubricating oil pressure and scavenging pumps as required; fuel oil service pump, regulating and overspeed governors, alarm and shutdown devices, strainers, filters, coolers, thermostatic control valves, mufflers, pyrometers, and manual flywheel barring device. Components such as coolers, filters, and mufflers need not be attached to the basic generator set.

3.8.2 *Lubricating system and oils.* Forced feed lubrication to all moving parts whether reciprocating or rotating, shall be supplied by a positive displacement rotary gear or reciprocating type pump. Drilled oil passages and securely fastenend pipes shall be used for distribution. Oil-cooling of pistons shall be required on all engines with a bmep of 75 psi or over. The engine shall operate satisfactorily with lubricating oil conforming to Specification MIL–L–9000. Metal edge strainer and absorbent type filters shall be supplied. The 30 kw set shall have a filter conforming to Specification F–F–351, type I, class 3, and shall be arranged as shown on Drawing B–204. The 60 kw set shall have two such filters as referenced above placed in parallel. The 100 and 150 kw sets shall each be equipped with two filter elements conforming to Specification MIL–F–3623 and arranged as shown on Drawing B–204. The 200, 250, and 300 kw sets shall each have three filter elements, and the 400 and 500 kw sets shall have four filter elements conforming to MIL–F–3623. The lubricating oil system of all diesel engines shall be fitted with simplex type metal edge strainers (having a built-in relief valve), in accordance with Drawing B–204, except the use of strainers is optional in complete self-contained diesel engines with integral lubricating oil piping, coolers, filters, and related components, and all sizes below 100 bhp. Lubricating oil sump may be wet or dry type. An oil cooler shall be supplied with each engine except those smaller than 100 bhp, if engine manufacturer proves satisfactory operation without a cooler. Temperature of lubricating oil at the oil pump shall not exceed 230 degrees F nor be less than 130

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degrees F with the engine operating continuously at any load up to rated load under the conditions specified herein. Oil coolers may utilize the engine jacket water or sea water as the cooling medium. Coolers shall conform to Specification MIL–C–15730, type A.

3.8.3 *Cooling system.* The engine shall be equipped with a heat exchanger using sea water as a cooling medium to cool the circulating water. Temperature of the circulating water shall be regulated automatically by a thermostatically-controlled valve to maintain the temperature of the water leaving the engine at the manufacturer's recommended optimum value for any applied load within the engine rating. Cooler shall conform to Specification MIL–C–15730, type A. Fresh and sea water pumps shall be of the centrifugal type. Sea water pumps shall be of all-bronze or copper-nickel alloy construction. Drive arrangement shall be designed to prevent water leakage into the engine lubrication system.

3.8.4 *Fuel system.* In addition to internal injection control and distribution equipment, each engine shall be furnished with a positive displacement booster pump driven by the engine, bypass relief valve, and with duplex strainer and filter in acordance with Specification MIL–F–20627 and as shown on Drawing B–204. A hand pump for priming the engine fuel system shall be provided where required. Means shall be provided to prevent fuel oil which may leak from the injectors, fuel pumps, or connections thereto, from entering the lubricating oil system. Constant pressure mechanical fuel injection (rail type) will not be acceptable.

3.8.5 *Cranking system.* All engines used as prime movers for generator sets rated 150 kw and over shall be arranged for compressed air or hydraulic cranking, as specified. Engines for generator sets rated less than 150 kw shall be arranged for electric, compressed air, or hydraulic cranking, as specified.

3.8.5.1 *Electric cranking.* Electric cranking shall be by cranking motor and gear-engaging mechanism. The cranking motor shall be the heavy-duty type with adequate capacity to crank the engine repeatedly or for a sustained operation of several minutes. The drive mechanism for engaging the cranking motor with the engine flywheel shall be designed to inherently engage and release without binding. Batteries shall not be furnished, but proper size shall be recommended, as chosen from Specification MIL–B–15072. A static type battery charger conforming to Specification MIL–C–19420 shall be furnished. The battery charger shall operate from the generator terminal voltage.

3.8.5.2 *Compressed air cranking.* Air cranking may be by direct admission to the cylinder or by separate air cranking motor. The necessary controls and appurtenances to provide a complete and operable system, when connected to a source of compressed air, shall be provided.

3.8.5.2.1 *Air compressor.* The number of compressors supplied shall be as specified. When one compressor is specified, it shall be of the combination electric motor and diesel-engine-driven type. When two compressors are specified, one shall be driven by an electric motor and the other by a diesel engine. Each compressor shall be sized to charge the two storage containers to working pressure within one hour. Compressors shall be of the multistage water- or air-cooled type, with intercoolers, capable of operating against whatever discharge pressure is needed for starting the engine, but in no case less than 250 psi gage. Motor driven compressors shall be furnished with automatic pressure control switches; diesel-engine-driven compressors shall be provided with pressure operated cutoff switches. Electric motors shall conform to AIEE Standard No. 45. Diesel engines shall be radiator cooled, electric motor cranked with provisions for hand cranking.

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3.8.5.2.2 *Storage tanks.* Two storage tanks shall be supplied with each compressor, each tank with sufficient capacity to provide 10 successive starts for the largest engine included in the contract without replenishment of air. Tank construction shall meet the pressure container requirements of the ABS Rules for Building and Classing Steel Vessels.

3.8.5.3 *Hydraulic cranking.* The hydraulic cranking system shall consist of a hydraulic cranking motor, one or more piston type accumulators with sufficient capacity for not less than three consecutive engine starts in 32 degrees F ambient temperature without recharging, an oil reservoir, an engine-driven pump, an emergency hand operated pump, a pressure gage, and all the connecting lines, valves, and fittings to make a complete cranking installation.

3.8.6 *Alarm and shutdown system.* An alarm system shall be provided which shall include an electrically operated 6-inch gong or horn actuated by a thermostatic switch arranged to indicate high jacket water temperature, and by a pressure switch installed to indicate low lubricating oil pressure. Temperature and pressure switches shall conform to Specification MIL–S–16032. The system shall operate at 115 volts, 60 cycle, single phase ac. Electric cable shall conform to Specification MIL–C–2194, type SGA, and shall be installed in an oil protected location. The shutdown system shall be mechanically operated, arranged to cutoff fuel to the engine when the lubricating oil pressure drops to a dangerous value, or when the jacket water temperature rises more than 10 degrees F above the alarm value. System shall be of the type which will not permit engine operation when actuating element fails.

3.8.7 *Air coolers.* When an air cooler is required for satisfactory performance of a supercharged four cycle engine, the air cooler shall be designed for an ambient air temperature of 125 degrees F. When sea water is used as the coolant, the design shall be based on an entering sea water temperature of 85 degrees F. Sea water coolers shall be constructed of materials equivalent to those in Specification MIL–C–15730, type A coolers.

3.8.8 *Governors.* Each engine shall be equipped with a regulating governor of the hydraulic type, and an independently driven overspeed governor.

3.8.8.1 *Speed governing system.* The speed governing system shall conform to AIEE Specification No. 606, section III, including the optional requirements for the motor actuated speed changer, speed regulation changer, and load limit changer. A 115-volt, split-field, series wound, reversible motor shall be employed on the motor actuated speed changer.

3.8.8.2 *Overspeed governor.* The overspeed governor shall be the manual reset, trip type, arranged to cutoff fuel completely to the engine, and set to prevent the engine from exceeding rated speed by more than 15 percent.

3.8.9 *Thermometers, mufflers, and pyrometers.* Thermometers shall conform to Specification MIL–T–656; mufflers to Specification MIL–M–15337; and pyrometers to Specification MIL–P–16304. Pyrometers shall include the necessary thermocouples and selector switch for the measurement of the exhaust temperature from each individual combustion chamber.

3.8.10 *Instrumentation.* An instrument panel shall be furnished and equipped with the instruments specified below, of the back connected, flush mounted type. The panel shall be located so that the instruments can be easily read by the operator and mounted to prevent damage by vibration. Instruments other than those specified below which are required for normal operation of the engine shall be furnished.

(a) Lubricating oil pressure gage.

(b) Low pressure fuel oil system gage.

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(c) Tachometer with dial graduated from 0 to 125 percent of rated engine speed.

(d) Hour meter, having a totalizing mechanism of not less than 9999 hours, to register the number of hours of engine operation at normal governed speed.

(e) Cooling liquid temperature indicator.

(f) Pyrometers.

3.9 **Generator.** Generators shall be in general accordance with AIEE Standard No. 45. Equipment furnished with the generator shall include an exciter, automatic voltage regulator, and exciter field rheostat. In general, all equipment shall be of the drip-proof protective type.

3.9.1 *Design.* Generators shall be of the rotating field, salient pole type. Stator windings shall be star connected, and all windings shall be ungrounded. Amortisseur winding shall be provided. Rated output voltage shall be 450 volts, 3 phase, at a rated frequency of 60 cycles per second. The generator shall be designed for continuous operation at rated voltage and kw at a power factor of 0.80 lagging in an ambient temperature of 50 degrees centigrade (C); temperature rises of various parts shall conform to AIEE Standard No. 45.

3.9.2 *Performance.* The generator shall be capable of carrying 150 percent rated current for 5 minutes, at rated voltage and frequency, and a power factor of 0.50 lagging. The generator shall be capable of withstanding, with constant field excitation, a 3 phase short circuit at the terminals for a period of 2 minutes, and a single phase short circuit between any two terminals for a period of 2 minutes. With a single phase load of 15 percent of rated current at any power factor between rated and unity, and with no other load on the generator, the variation in voltages between phases shall not exceed 5 percent of rated voltage. During the phase unbalance heating test described hereinafter, the temperature rises of the various parts of the generator shall not exceed the specified values for full load rating. The generator shall be capable of operation at 125 percent of rated speed without an appreciable increase in vibration over operation at rated speed and without injury to generator or exciter. The inherent voltage regulation of the generator, operating at constant excitation and frequency, shall not exceed 40 percent. The subtransient reactance of generators rated below 200 kw shall have a minimum of 0.10 per unit and generators rated at 200 kw and above shall have a minimum of 0.18 per unit. The deviation factor of the open circuit terminal voltage shall not exceed 10 percent on any phase.

3.9.3 *Construction.* Generators and exciters shall be of substantial construction throughout, with major consideration being given to ready disassembly and replacement of parts in the field. Jackscrews and dowel pins or fitted bolts shall be provided as required for satisfactory adjustment of alinement with the primer mover. Exciters shall be of the direct-connected, or of the belt-connected, top-mounted type. Generators may be one or two bearing type. Ball or sleeve bearings may be furnished on 30 and 60 kw generators having a flexible coupling between the engine and generator. Sleeve bearings shall be required on all other generators. The lubricating system shall be designed to prevent oil or grease from spilling out of its reservoir under the described inclined operation and rolling conditions.

3.9.3.1 *Frame.* The generator frame shall be designed and constructed to maintain permanent correct alinement of all parts which it supports. Openings with removable covers in the end bracket at the collector-ring end shall be provided to give easy access to the brushes, brush rigging, and generator collector-rings and to the exciter commutator.

3.9.3.2 *Rotor.* The rotor shall be designed for safe operation up to 25 percent overspeed and balanced dynamically to such a degree

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that at the rated speed of the rotor, the remaining unbalanced centrifugal force, in ounce-inches, in either of the planes of correction, shall not exceed 5,600 times the weight of the rotor divided by the square of the rated rpm. Balancing shall be affected by drilling out material, by the use of noncorrodible balancing weights which are securely locked; by rivets; by lead or babbitt metal carried in a receiver in such a manner as to preclude the possibility of its breaking loose; by steel weights anchored in balance grooves; or by steel weights securely welded in place.

3.9.3.3 *Sleeve bearings.* Pedestal mounted sleeve bearings shall be of the self-alining type. Bracket mounted bearings need not be self-alining. The bearings shall be split in such a manner as to permit removal and replacement without taking the rotor out of the frame, removing the coupling from the driving end, or removing the bearing brackets from the generator. For generators smaller than 100 kw, the bearings need not be split if the bearing shell and oil seal are readily removable without the necessity of removing the bearing bracket, and if the bearing shell is removable without removal of overhung exciter, armature, or coupling. Sleeve bearings shall be lubricated by a ring, disk, or forced-feed lubrication system. Hinged snap oil rings shall not be permitted. The bearing housing shall be arranged to prevent the rings from being displaced when the dock is trimmed or listed, or rolling, and to prevent escape of oil along the shaft. Felt and friction type seals are not acceptable for the latter purpose. Provision shall be made for observation of oil rings, and for insertion of a thermometer conforming to Specification MIL–T–656, type B, to determine the temperature of the bearing and shell while the generator is in operation. Readily accessible ⅜ inch minimum diameter opening or standpipe for filling the bearing together with means for indicating the oil level, shall be provided. The observation and filling openings shall be made oiltight by a screwed plug, a cover secured by screws, or equivalent means.

3.9.3.4 *Ball bearings.* Ball bearings shall be of the single shielded type conforming to Specification FF–B–171, and where grease is used as a lubricant shall be provided with readily accessible compression grease cups conforming to Standard MS 35758 or MS 35759. Drain plugs for removal of grease shall be provided at the bottom of the bearing housing.

3.9.3.5 *Brushes.* The brushes shall conform to Specification MIL–B–3743, grade E or H, unless specifically approved otherwise.

3.9.3.6 *Exciter.* The exciter shall be stabilized for the specific generator, shall be shunt-wound, self-excited for 125 rated volts, and designed for satisfactory operation with a voltage regulator of the direct-acting rheostatic type. It shall have a capacity sufficient to meet the overload requirements of the generator, except that it shall have a minimum capacity of 120 percent of that required for excitation of its alternator at rated load and power factor. The speed of the exciter shall not be over 1,800 rpm. Exciter field rheostat shall be in accordance with Specification MIL–R–15109, form P, with class C insulation.

3.9.3.7 *Automatic voltage regulator.* Automatic voltage regulator shall be of the direct-acting rheostatic type, suitable for parallel operation and with individual droop compensation, conforming to Specification MIL–R–2729.

3.10 Radio interference suppression.

3.10.1 *For Navy use.* The generator set shall conform to Specification MIL–I–16910 for radiated and conducted radio interference. The requirement in Specification MIL–I–16910 excepting generator sets having power outputs in excess of 60 kva shall not apply.

3.10.2 *For Army use.* The generator set shall conform to the requirements of Specification MIL–I–11683 for radio interference suppression, except that the requirements

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covering radiated interference limits of engine generator units shall include the 60 kw capacity.

3.11 Fungus and moisture resistance. The generator set shall be treated to resist fungus and moisture as specified below.

(a) Electrical components such as switches, fuses, contacts, and heater elements shall not be treated. Other materials and components which are inherently fungus resistant or are protected by hermetic sealing need not be treated.

(b) Circuit elements, not covered above and which have a temperature rise of not more than 75 degrees F. when operating at full load, shall be coated with a fungus resistant varnish conforming to Specification MIL–V–173. Circuit elements include, but are not limietd to, cable, wire, switchboards, panelboards, terminal and junction blocks, junction boxes, capacitors, and coils.

(c) Circuit elements such as motor coils, generator windings, and similar electrical components, which have a temperature rise exceeding 75 degres F. when operating at full load, shall not be coated with fungitoxic compound. Instead, such components shall be treated in accordance with the methods specified in Specification MIL–E–917.

3.12 Lifting attachments. The generator set shall be provided with suitable lifting attachments to enable its being lifted in its normal position. The attachments shall be located so that, when hoisted, adequate clearance will exist between slings and all exterior parts of the generator set. When practicable, one attachment of sufficient capacity to carry at least 500 percent of the total lifted weight shall be used. When two or more slings are used, each attachment shall be of sufficient capacity to carry at least 250 percent of the weight being lifted. The attachments for multiple slings shall be designed so that the attached slings will converge at a point not to exceed 19 feet above the lowest extremity of the load when such limitation is possible. Lifting attachments shall be positionel in such direction that the lifting strain will be in line with the longitudinal axis (peripheral plane) of the eye of the attachment. The eye of each lifting attachment shall be not less than 3 inches in diameter. The members to which pad eyes are attached shall be of sufficient strength to withstand stresses in the amount and direction of pull specified for the pad eye. Where required, suitable reinforcements shall be used to meet the requirements specified herein. Examples of such reinforcements are, but are not confined to the following:

(a) Flat lifting pads shall be bent around the member to give a 90 degree effect and in such manner as to bear against the surfaces opposite to the direction of pull. Where it is not practicable to bend the pad around the member, a bolt or bolts of sufficient capacity shall pass through the pad and equipment members. In all cases, the pad shall be welded to the generator set members.

(b) Round stock pad eyes shall be bent around the member in such a manner as to bear against the generator set surfaces opposite to the direction of pull and shall be welded to the member. Where this is not practicable, the round stock shall be threaded and passed through the member and be bolted and welded on both sides of the member.

A diagram showing the lifting attachments shall be inscribed on a copper-base alloy plate securely fastened to the generator set with the lifting capacity of each attachment and

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the required length and size of each sling cable marked thereon. A silhouette of the generator set showing the center of gravity shall be provided.

3.13 Treatment and painting.

3.13.1 *Treatment.* Prior to painting, surfaces of the generator set to be painted shall be cleaned and treated in accordance with Specification MIL–T–704.

3.13.2 *Priming.* After cleaning and treatment, surfaces to be painted shall be given one coat of zinc chromate primer conforming to Specification JAN–P–735 applied to a minimum dry film thickness of one mil.

3.13.3 *Finishing.*

3.13.3.1 External surfaces shall be given a finish coat of light gray equipment enamel conforming to Specification MIL–E–15090, class 2, applied to a total minimum dry film thickness of three mils.

3.13.3.2 Internal surfaces of field frames shall be given one coat of black electrical-insulating varnish conforming to Specification MIL–V–1137, type M. Other internal surfaces normally painted shall be given one finish coat of black enamel conforming to Specification TT–E–489.

3.14 Stenciling. The gross weight of each complete generator set shall be stenciled on each side of the set in such a manner as to be readily discernible to dock and military personnel. Stenciling shall conform to Specification MIL–T–704.

3.15 Identification marking. A copper-base alloy identification plate conforming to Standard MIL–STD–130, marked with the information specified in AIEE Standard No. 45 shall be permanently affixed to the generator set in a conspicuous place.

3.16 Instruction plates. The generator set shall be provided with instruction plates suitably located, describing any special or important instructions or procedures to be followed in operating or servicing the set and shall caution against opening a hot crankcase within 10 minutes after shutdown.

3.17 Technical manuals. Unless otherwise specified, the contractor shall furnish four copies of technical manuals. Technical manuals shall conform to Specification MIL–M–15071, type III. All information contained in the manual must be strictly applicable to the specified generator set. The following components shall be covered in the technical manual:

Diesel engine and associated accessories.

Generator.

Exciter.

Voltage regulator.

Field rheostat.

Starting equipment.

Alarm equipment.

Governor motor.

Special devices.

The technical manual shall include a list of approved plans, indicating the manufacturer's plan number, and an approved list of repair parts.

3.18 On board repair parts. A list of repair parts, together with prices, shall be furnished as soon as possible after award of contract. The number of parts shall be in general accordance with table II, based on the number of the same size generator sets intended for each dock. After review of the list, repair parts, if procured, will be purchased as a separate item. Information as to distribution of the sets on each dock shall be obtained from the contracting officer.

3.19 Maintenance tools. All special tools required for assembly, disassembly, and adjustment of the generator set and its parts, shall be furnished.

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TABLE II. *On Board Repair Parts for Generator Sets.*

Name of part	No. per dock per engine size	
	Less than 5 engines	5 or more engines
Engine cylinder head, complete with valves, springs, etc.	1	1
Piston, complete with wrist pin and bushing (no rings)	1	1
Connecting rod	1	1
Parts liable to damage or wear for piston cooling arrangement of one cylinder, set	1	2
Valves for one cylinder, complete with springs, etc., sets, percent	20	25
Cylinder liner	1	2
Packing, etc., for installation of each cylinder liner, sets, percent	30	20
Piston rings, percent	30	20
Main bearing shells (complete), percent	30	20
Crankpin bearing shells (complete), percent	30	20
Main bearing shells (complete), percent	10	10
Crankpin bearing assembly bolts and nuts, sets, percent	10	10
Crankshaft bearings, sets	1	2
Gaskets, sets (at least one of each size), percent	50	25
Bolts or studs for one cylinder cover, sets, percent	10	10
Templates and gages for alining main bearing sets	1	2
Fuel injector pumps	1	10 percent
Fuel injector nozzles, percent	25	15
Fuel injector springs and other wearing parts, percent	50	30
Fuel oil lines between injection pump and injectors, sets	1	10 percent
Fuel oil pump, complete	1	1
Fuel oil pump, set of repair parts	2	
Lube oil pump, complete	1	2
Filter and strainer elements, sets	1	15 percent
Air starting valves, percent	15	10
Pyrometer thermocouple assemblies, percent	15	10
Scavenging blower rotor, complete	1	2
Turbocharger rotor, complete	1	2
Regulating governor, complete	1	1
Regulating governor, set of repair parts	1	2
Overspeed governor, complete	1	1
Overspeed governor, set of repair parts	1	2
Jacket water pump, complete	1	1
Jacket water pump, set of repair parts	2	3
Saltwater circulating pump, (engine driven) complete	1	2
Saltwater circulating pump (engine driven) set of repair parts	2	4
Engine-to-generator coupling bolts, sets	1	10 percent
Each size and type of flexible connector between dock source and engine for fuel and water	1	10 percent
Generator and exciter repair items:		
Bearings or bearing linings (sleeve bearings shall be complete), sets	1	25 percent
Oil gages for sleeve bearings (each type used)	1	10 percent
Bearing lubricant seals (sleeve bearing generators only), sets	2	25 percent
Brushes, sets	3	50 percent
Brush rigging insulation, complete sets	1	10 percent
Brush holders, complete with springs	1	10 percent
Brush holder springs, set	1	25 percent
Field coils, complete set (if wound directly on pole, including pole)	1	10 percent
Armature winding, slot insulation and wedges, complete set	1	10 percent
Field rheostat, complete	1	1
Automatic voltage regulator	1	10 percent
Synchronizing motor	1	1

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3.20 Items to be furnished with prime equipment. Maintenance tools and accessories shall be included with the shipment of the prime equipment, unless written approval for shipment of the prime equipment without such items, has been received from the activity concerned.

3.21 Drawings. Prints containing information necessary to demonstrate conformance to the requirements, and information necessary for installation, and mathematical analysis shall be forwarded to the contracting officer for approval within 45 days after receiving the contract. After prints have been approved, one complete set of reproducible drawings shall be provided. The tracings shall be capable of producing legible, undistorted prints, shall have all references to contract numbers and drydocks numbers deleted therefrom, and shall include and index of tracings. Once such reproducibles are on file in the procuring agency for future purchases of identical, or nearly identical, equipment, the manufacturer will be required to supply only certification that the units are identical, or for nearly identical equipment, only such additional reproducibles as are necessary to illustrate the differences. Where listing of material composition of various parts is required, the material designation shall comply with the applicable Military or Federal Specification. Information required on the drawings shall include:

(a) Dimensioned outline plan of the complete generator set; weight of subbase, engine and generator; location of center of gravity of the engine and of the complete set; foundation bolting dimensions; and clearance required for renewal of cylinder heads, liners, pistons, and accessory equipment.

(b) Details of the subbase.

(c) Details of flexible coupling, including material designation of parts.

(d) Torsiograph of complete unit.

(e) Details of the engine, including:

1. Cross-sectional views showing internal assembly of engine in sufficient detail to show clearly how all removable parts are assembled and fastened.
2. Wet and dry weights of engine and its attached accessories.
3. Bmep and bhp to supply the generator rating.
4. Maximum continuous bhp and bmep.
5. Blower capacity in free air per minute (2 cycle engines).
6. Diagrammatic sketches of various piping systems indicating pipe sizes, types and sizes of suction and discharge connections, relative locations of coolers, valves, filters, thermometers, and gages recommended by manufacturer.
7. Wiring diagrams of alarm and starting systems.
8. Assembly and sectional drawings, including material designation of parts, for accessory equipment such as coolers, sea water and jacket water pumps, lubricating oil and fuel pumps, etc.

(f) Piping diagrams, showing pump capacities, normal working pressures and flow of gallons per minute through various parts of the piping systems.

(g) Details of the generator, including:

1. Performance table, consisting

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at least of normal and overload kw and current ratings, voltage rating, power factor, guaranteed efficiency, and required direct current excitation.

2. An assembly end and side view, showing all parts of the generator identified with piece numbers in the list of materials. The assembly side view shall show a longitudinal section of the generator above the centerline. The assembly end view shall show a tranverse quarter section above the centerline. The drawings shall show clearly how all removable parts are assembled and fastened, method of bringing out and protecting terminal cables from chafiing and strain, direction of air flow and overall dimensions, including space to remove rotor.

3. A detailed working drawing of the shaft, including material composition, dimensions, and tolerances.

4. Enlarged section through field coils showing methods and type of insulation, and field coil data, including connection diagram, number of turns, size of conductor and developed length, resistance at 25 degrees C, weight of conductor per coil, arrangement of turns by layers, and insulation and varnish data.

5. Stator winding data, including winding diagram, number of slots, number of coils per slot, number of turns per coil, pitch of coil, size of conductor and developed length, weight of conductor wire per coil, resistance of winding at 25 degrees C, circuits and phase number out of frame, and insulation and varnish data.

6. Details of sleeve bearing shell and liner, including dimensions, material composition of lining, ring or disk details, and method of filling and draining bearing.

7. A note identifying replacement ball bearings by the Government standard bearing number, and identifying the applicable standard bearing puller, where applicable.

8. Half section drawing through the exciter showing how all removable parts are assembled and fastened, and details of drive from the generator.

9. Number of brushes, dimensions of brush, Government grade and manufacturer's name and grade of brush, for both the generator and exciter brushes.

10. Data obtained from tests of first generator, including:
    a. Efficiency (rated power factor) at full, ¾, and ½ load.
    b. Breakdown of full load and rated speed losses in kw, into armature I^2R loss, field I^2R loss, core loss, friction and windage loss, brush contact loss, stray load losses.
    c. Resistance of stator and rotor at 75 degrees C.

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d. Heat run temperature rise on all major parts of generator, including method of measurement, and ambient temperature.

e. Open circuit saturation values.

f. Synchronous impedance values.

g. Voltage and current balance values.

h. Overspeed rpm attained, and behavior of generator.

i. Short-circuit ratio, subtransient reactance and time constant, transient reactance and time constant, and negative-sequence impedance.

j. Dielectric strength of stator and rotor.

k. Insulation resistance hot and cold, with recorded temperature and humidity for each case.

l. Actual weights of rotor and complete generator.

m. Air gap measurements.

n. Actual dynamic unbalance in ounce-inches.

(h) Master plan of automatic voltage regulator.

(i) Master plan of exciter field rheostat.

3.22 **Workmanship.** Workmanship shall be of the highest grade throughout and in accordance with good commercial practice for this type of equipment.

3.22.1 *Steel fabrication.* Steel used in the fabrication of equipment shall be free from kinks and sharp bends. The straightening of material shall be done by methods that will not cause injury to the metal. Shearing and chipping shall be done neatly and accurately. Corners shall be square and true. Flame cutting, using a tip suitable for the thickness of metal, may be employed instead of shearing or sawing. Re-entrant cuts shall be made in the best possible manner. All bends of a major character shall be made with controlled means in order to insure uniformity of size and shape. Precautions shall be taken to avoid overheating, and heated metal shall be allowed to cool slowly.

3.22.2 *Bolted connections.* Bolt holes shall be accurately punched or drilled and shall have the burrs removed. Washers or lockwashers shall be provided in accordance with good commercial practice, and all bolts, nuts, and screws shall be tight.

3.22.3 *Riveted connections.* Rivet holes shall be accurately punched or drilled and shall have the burrs removed. Rivets shall be driven with pressure tools and shall completely fill the holes. Rivet heads, when not countersunk or flattened, shall be of approved shape and of uniform size for the same diameter of rivet. Rivet heads shall be full, neatly made, concentric with the rivet holes, and in full contact with the surface of the member.

3.22.4 *Welding.* The surface of parts to be welded shall be free from rust, scale, paint, grease, or other foreign matter. Spot, tack, or intermitent welds for strength will not be permitted. Weld penetration shall be such as to provide transference of maximum design stress through the base metal juncture. Fillet welds shall be provided when necessary to reduce stress concentration. Before assigning any welder to manual welding work, the manufacturer shall certify to the contracting officer that the welder has passed the qualification tests as prescribed by either the American Welding Society or the American Society of Mechanical Engineers for the type of welding operation to be performed.

3.22.5 *Machine work.* Tolerances and gages for metal fits shall conform to the standards of good commercial practice.

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3.22.6 *Castings.* Castings shall be sound and free from patching, misplaced coring, warping, or other defects which might render the casting unsound for use.

4. QUALITY ASSURANCE PROVISIONS

4.1 The supplier is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified, all inspections shall be conducted or performed at a site designated by the supplier. The supplier may utilize his own or any other inspection facilities or services acceptable to the Government. Inspection records of the examinations and tests shall be kept complete and available to the Government for the period of time specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification, where such inspections are deemed necessary to assure that the products conform to the prescribed requirements.

4.2 **Classification of inspection requirements.**

(a) Preproduction model inspection.

(b) Production unit inspection.

(c) Preparation for delivery inspection.

4.3 **Preproduction model inspection.** Preproduction inspection shall be performed on one complete generator set designated as the preproduction model. The inspection shall include, but shall not necessarily be limited to the examinations specified in 4.5 and the tests specified in table III and 4.6.4.

4.4 **Production unit inspection.** All production units shall be inspected prior to submission to the Government for acceptance and delivery. The inspection shall include but shall not necessarily be limited to the examinations specified in 4.5 and the tests specified for duplicate units in table III.

4.5 **Examination.** Each generator set shall be subjected to visual and tactile examination during fabrication, manufacture, assembly, or after completion, as appropriate or timely. The purpose of the examination shall be to determine whether or not the generator set complies with the applicable requirements of this specification and with the provisions of the contract that pertain. The following list of potential defects is provided for guidance and shall not be considered as all-inclusive or final. The Government reserves the right to include any other defects that arise or are found during the inspection processes; or that otherwise develop prior to Government acceptance. Any generator set found to have one or more defects shall be rejected:

Examine	Requirement	Defects
Material	3.2	Material used in fabrication or manufacture of any component, part, attachment, or accessory not as specified or not suited to the purpose.
Safety	3.5	Any unauthorized deviation or departure from specified regulations or requirements. Any existing condition or arrangement which presents or may develop into a hazard to the safety of operating or attending personnel.
Design	3.7 through 3.7.2	One or more details not in accordance with specified requirements.
Engine	3.8	Any unauthorized deviation from table I.
Accessories	3.8.1	One or more missing, inadequate, or incompatible.
Lubricating system and oils	3.8.2	Not as specified.
Cooling system	3.8.3	Functionally inadequate.
Cranking system	3.8.5 through 3.8.5.3	One or more details not according to specified requirements.

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Examine	Requirement	Defects
Alarm and shut-down system	3.8.6	Missing, incomplete, or inadequate.
Air cooler	3.8.7	Material not as specified.
Governors	3.8.8 through 3.8.8.2	Type and system not as specified.
Thermometers, mufflers, and pyrometers	3.8.9	Not as specified.
Instrumentation	3.8.10	One or more specified instruments missing or unsuitable.
Generator	3.9 through 3.9.3.7	Any unauthorized deviation or departure from the specified design. Any evidence of incapability of performing according to specified details. One or more unauthorized deviations from construction requirements. Method of balancing rotor not acceptable. Sleeve and ball bearings not as specified. Exciter and voltage regulator inadequate, incompatible, or otherwise not according to requirement.
Radio interference suppresion	3.10.1 3.10.2	Not as specified.
Lifting attachments	3.12	Design or material not as specified, unsuitably or awkwardly located, or improperly, insecurely, or unsafely attached.
Treatment and painting	3.13.1 through 3.13.3.2	Surfaces to be painted not treated as specified prior to application of primer. Type or color and thickness of finish not as specified. Finish blistered, peeled, chipped, scratched, oversprayed, thin, bare spots, or otherwise impaired to the extent that appearance or serviceability are affected. Internal surfaces not coated as specified.
Stenciling	3.14	Missing or not according to specified requirement.
Identification and instruction plates	3.15 3.16	Missing, or not of type, size, or material specified. Not in specified location; not affixed in prescribed manner, or insecurely fastened. Data incomplete, illegible, or erroneous.
Technical manuals	3.17	Missing, incomplete, or otherwise not as specified.
Repair parts	3.18	Any unauthorized deviation from, or circumvention of the detailed requirements as specified.
Drawings	3.21	Not as specified.
Workmanship	3.22	Not as specified.
Steel fabrication	3.22.1	One or more steel members bent, twisted, or kinked at points other than where required by accepted design or configuration. Evidence of overheating or poor straightening methods. Rough, sharp, or uneven edges.
Bolted and riveted conections	3.22.2 3.22.3	One or more bolts or rivets missing, broken, bent, loose, or not of specified type or size. One or more holes rough, burred, oversize, or out-of-round.
Welding	3.22.4	Poor appearance, incomplete, porous, sparse, flux inclusions, excessive spatter, undercuts, or lack of fusion with parent metal or between parts.
Machine work	3.22.5	Surfaces rough, pitted, uneven, or tool marked. Tolerances on fit-ups not according to specified requirements.
Castings	3.22.6	Blowholes, patches, warps, excessive fins, or other similar or characteristic defects impairing the function, operability, reliability, or appearance of the piece, part, or assembly.

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4.6 Tests. Tests shall include special paralleling tests on the floating drydock with factory supervisor present after installation of the generator set; special tests on the first complete set of each type as well as the first engine and generator; and routine tests on duplicate sets, engines, and generators. Tests shall be conducted as specified in table III.

Table III. *Outline of Tests*

Description of test	Assembled set		Engine		Generator	
	Preproduction model	Duplicate units	Preproduction model	Duplicate units	Preproduction model	Duplicate units
Dynamic unbalance					X	X
Stator and rotor resistance					X	
Voltage and current balance					X	X
Cold starting			X			
Normal starting			X	X		
Run-in period	X			X		
Endurance test			X			
Overload test			X	X		
Post trial examination			X			
Torsiograph	X					
Inclination	X					
Governor	X	X				
Voltage regulation	X	X				
Generator airgap measurement	X	X				
Overspeed govenor setting	X	X				
Weight measurement	X					
Overspeed					X	X
Dielectric					X	X
No load saturation curve data					X	
Polarity of field coils						X
Efficiency and measurement of losses					X	
3 phase short circuit for 120 seconds					X	
Measurement of synchronous impedance, transient and subtransient reactances and negative sequence impedance					X	
Short circuit characteristic data					X	
Determination of voltage droop					X	
15 percent phase unbalance heating test					X	

4.6.1 *Test procedure, generator.* Test procedures not listed below or previously specified shall be as recommended by the AIEE Standard No. 45.

4.6.1.1 *Dynamic unbalance.* The actual unbalance shall be measured and checked against the maximum permissible value. Both values shall be included in the test report of the first generator.

4.6.1.2 *Stator and rotor resistance.* The direct current resistance of the stator and rotor windings and the temperature at which they are measured shall be taken and recorded.

4.6.1.3 *Voltage and current balance.* The voltage balance test shall consist of simultaneous measurements of voltage in each phase at an excitation producing rated voltage in at least one phase. Voltage between each phase, and the field current and voltage shall be recorded. Data may be taken at no load.

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The current balance test shall consist of simultaneous measurements of current in each phase at an excitation which produces rated current in at least one phase. The current per phase, field volts, and field current shall be recorded. Data may be taken with generator terminals short circuited and sufficient excitation to produce rated current.

4.6.1.4 *Phase unbalanced heating test.* The generator shall be loaded with a single phase load of 15 percent of continuous rated current simultaneously with balanced three phase load, the total load to be at not less than rated power factor, and to be such that the current in two phases shall be approximately, but not to exceed, continuous rated current. The single phase load shall be at unity power factor. The test shall be continued until steady final temperatures are reached in all parts of the generator.

4.6.2 *Test procedure, engine.* Procedures for test applying to the engines only, shall be as outlined below. Where not definitely described, data recorded during the tests shall be in general accordance with DEMA Standard Practices for Stationary Diesel and Gas Engines.

4.6.2.1 *Cold starting.* Prior to each start, the engine shall be preconditioned to a stabilized temperature not higher than 35 degrees F. Preconditioning may be accomplished by circulating a cold liquid through the engine, by the use of a controlled temperature room, or by any suitable method that will condition the engine to the specified temperature. The engine shall fire and continue to operate within 10 seconds after the cranking mechanism has been set in operation. For electric cranking, record the power required for each of three starts. For air cranking, the quantity of free air at 14.7 psi and 68 degrees F. necessary for starting shall be determined by averaging three such starts. For hydraulic cranking, the accumulators shall be pumped to full operating pressure, the engine-driven pump shall be disconnected or bypassed, and the engine shall be started three consecutive times to determine conformance to 3.8.5.3.

4.6.2.2 *Normal starting.* The engine shall be started at least three times in an ambient temperature of approximately 68 degrees F. The engine shall fire and continue to operate within five seconds after the cranking mechanism is engaged. For electric cranking, determine the maximum power required to start the engine. For air cranking, determine the amount of free air at 68 degrees F. and 14.7 psi that is required for 10 successive starts and the minimum air pressure at which the engine will start within the five second time limit. For hydraulic cranking, determine the minimum hydraulic pressure at which the engine will start within the five second time limit. Regardless of the type of cranking employed, the minimum rpm at which the engine will start shall be determined. Duplicate units shall be tested only to determine that the starting time is within the specified limit and that the minimum starting rpm is within 15 percent of the value obtained for the first engine.

4.6.2.3 *Run-in period.* The engine shall be subjected to a run-in and conditioning period in accordance with the manufacturer's standard practice. During the test period, observe general operation of the engine and make adjustments as required.

4.6.2.4 *Endurance test.* This test shall be a continuous 12-hour nonstop test, and only minor interruptions which in the opinion of the Government representative do not influence the ability of the engine to withstand such a test, will be permitted. In case of a more serious breakdown, the cause of it shall be corrected satisfactorily and the complete test repeated. The engine shall be operated at rated speed and at ½ load for 2 hours, ¾ load for 2 hours, and rated load for 8 hours. During the test, record the exhaust temperatures and the inlet and outlet temperatures of the cooling water and lubricating oil every 30 minutes. Obtain fuel consumption for the ½, ¾, and rated load operation.

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4.6.2.5 *Overload test.* Immediately following the endurance test, apply 110 percent of rated load for two hours, observing mechanical balance, noise, vibration, and temperature rise. Duplicate engines shall be tested at 110 percent of rated load for 15 minutes.

4.6.2.6 *Post trial examination.* Upon completion of the above tests, the engine shall be subjected to a post trial examination. For this purpose, at least one cylinder shall be opened, the piston pulled, and the cylinder bore, piston, and piston rings examined. Wrist pin and crankpin bearings shall also be examined, and the center main bearing shell rolled out after the crankshaft deflection has been measured and recorded. Gear train covers shall be removed and the condition of gears, bearings, or bushings shall be determined.

4.6.3 *Test procedure for complete generator set tests.* The test procedure shall be as outlined below:

(a) *Parallel operation.* A continuous, two-hour test with all generators synchronized, shall be conducted on the dock after installation of the generator sets or when specified, with identical generators at place of manufacture. With each generator at rated voltage, and at approximately 20 percent load, and at rated power factor, adjust the governors and voltage reactive droop compensation for equal kw and kvar load division. No further adjustment of governors, voltage regulator or rheostats shall be made for the duration of the test. Vary the load from 20 to 100 percent of the combined ratings of the generators, at about rated power factor, in four approximately equal steps. Maintain 75 to 100 percent load for 90 minutes. Then reduce load in four approximately equal steps to 20 percent and vary again from 20 to 100 percent and back to 20 percent. At each load point, record speed, voltage, current, kw load, kvar load, and field current of each generator.

(b) Torsiograph the generator set to determine location of critical speeds and their maximum amplitude. Torsiograph records shall be suitably analyzed before forwarding to procuring agency.

(c) *Inclination test.* Operate the assembled set inclined 5 degrees from the horizontal in longitudinal direction for a minimum of 15 minutes. Operate the assembled set inclined 15 degrees in the transverse direction for a minimum of 15 minutes. Run at rated speed with excitation. Ascertain that mechanical balance and lubrication are satisfactory, with no pounding or grinding at the bearings, or leakage of lubricant from the bearings.

(d) *Governor test.* Records of speed changes due to sudden load variations from full load to no load by 25, 50, 75, and 100 percent increments, and by similar increments, from no load to full load shall be obtained. The steady state speed variation between sudden load changes shall also be shown.

(e) *Voltage regulation.* The generator shall be tested with its automatic voltage regulator to determine conformance to 3.7.2 and 3.9.3.7.

(f) Generator airgap measurements shall be measured in at least 8 places about 45 mechanical degrees apart including one measurement at the bottom.

(g) *Overspeed governor setting.* Determine by test the actual speed at

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which overspeed governor operates.

4.6.4 *Radio interference suppression tests.*

4.6.4.1 *For Navy use.* Tests shall be conducted on the preproduction model for radiated and conducted interference to determine conformance to 3.10.1. All testing equipment, instruments, personnel for making the test, and necessary facilities shall be furnished by the contractor. The test location shall be reasonably free from interference. Duplicate generator sets shall be inspected for conformance to the approved preproduction model.

4.6.4.2 *For Army use.*

4.6.4.2.1 *Preproduction model.* The preproduction model of each separate design and rating shall be tested by the Government for conformance to 3.10.2, after which, if the interference suppression system is satisfactory, the contracting officer will furnish to the contractor a report detailing the suppression system to be followed on all subsequent units of production of identical design. Units of greater than 60 kw rating will not be tested for radiation, but will be tested and inspected for conformance to other requirements of the applicable specification.

4.6.4.2.2 *Production testing.* The contractor shall visually inspect each unit to determine conformance to the contracting officer's report on the approved suppression system.

4.7 **Inspection of preparation for delivery.** The preservation, packaging, packing, and marking of the generator set shall be inspected to determine compliance with section 5.

5. PREPARATION FOR DELIVERY

5.1 **Preservation and packaging.** The preservation and packaging shall be level A or C, as specified.

5.1.1 *Level A.*

5.1.1.1 *Disassembly and matchmarking.* Disassembly shall be the minimum necessary to safeguard parts known to be subject to damage or loss and to accomplish reduction in cube. Bolts, nuts, screws, pins, and washers removed shall be reinstalled in one of the mating parts and secured to prevent their loss. When necessary, to facilitate reassembly, parts removed and mating parts shall be matchmarked. Matchmarking shall be printed on shipping tags conforming to Specification UU–T–81, type A. The printed tags shall be waterproofed in accordance with Standard MIL–STD–129.

5.1.1.2 *Preservatives.* Preservatices shall conform to the applicable specifications listed in, and shall be applied in accordance with Specification MIL–P–116.

5.1.1.3 *Unpainted surfaces.* Unpainted exterior ferrous metal surfaces including threaded surfaces and surfaces exposed by disassembly shall be coated with type P–1 preservative.

5.1.1.4 *Generator and exciter.* The generator and exciter shall be preserved in accordance with Specification MIL–P–16298 and shall be packaged in accordance with the alternate method described therein.

5.1.1.5 *Engines.* Engines, component parts, and accesories shall be preserved in accordance with Specification MIL–E–10062 and the following:

5.1.1.5.1 *Exhaust pipes and mufflers.* After completion of engine preservation, unpainted exterior surfaces of exhaust pipes and mufflers shall be coated with type P–1 preservative, or heat-resisting paint conforming to Specification MIL–P–14105.

51.1.5.2 *Cranking systems.*

5.1.1.5.2.1 *Electric cranking systems.* The electric cranking motor shall be preserved

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and packaged as specified for generator and exciter.

5.1.1.5.2.2 *Compressed air cranking systems.* The internal surfaces of the air cranking systems shall be coated with type P-10, grade 2 preservative. The preservative shall be sprayed into air intakes of direct air systems, while applying air pressure at intakes, or of air motor systems, while air motors are operating, for a sufficient length of time to insure complete coverage of the internal parts and surfaces. Valves shall be closed and all openings into the system shall be sealed with tape conforming to Specification PPP-T-60, type II, class 1.

5.1.1.5.2.3 *Hydraulic cranking systems.* The hydraulic fluid supply reservoir shall be filled to the proper operating level with hydraulic fluid .The operating levers of the system shall be actuated to insure complete coverage of all interior parts with the fluid.

5.1.1.5.3 *Pumps.* Interior surfaces of pumps shall be coated with type P-10, grade 2 preservative in a manner to insure thorough coating of all interior parts and surfaces. Openings into the pumps shall be sealed with threaded metal or plastic caps or plugs, or barrier material conforming to Specification MIL-B-121, type I, grade A, class 2 secured with tape conforming to Specification PPP-T-60, type II, class 1. Flanged openings shall be covered with flange covers, inserting barrier material specified herein, between flange and cover.

5.1.1.5.4 *Air compressors.* The air compressors shall be preserved and packaged in accordance with level A of Specification MIL-C-3600.

5.1.1.6 *Drive belts and pulleys.* Belts shall either be released from tension or shall be completely removed. Unpainted faces of pulley grooves shall be coated with primer conforming to Specification TT-P-664. Belts removed shall be preserved in accordance with Specification MIL-P-116, method 1C-3.

5.1.1.7 *Disassembled components and repair parts.* Disassembled components and repair parts shall be preserved and packaged in accordance with Specification MIL-R-196, MIL-P-16298, MIL-E-17555, or in accordance with Specification MIL-P-116 for parts not specifically covered in Specification MIL-P-196, MIL-P-16298, or MIL-E-17555.

5.1.1.8 *Maintenance tools.* Maintenance tools shall be preserved and packaged in acordance with Specification MIL-H-15424.

5.1.1.9 *Technical publications.* Technical publications for each generator set shall be preserved in accordance with Specification MIL-P-116, method IC-1 or IC-3.

5.1.1.10 *Intermediate packaging.* Repair parts, maintenance tools, technical publications, and small dissembled components requiring additional protection from mechanical damage shall be intermediate packaged in close-fitting boxes conforming to Specification PPP-B-621 (class 1) or PPP-B-601 (domestic type). Contents shall be cushioned, blocked, and braced to prevent movement within the boxes. Strapping shall not be required.

5.1.2 *Level C.* The complete generator set shall be preserved and packaged in accordance with the supplier's standard practice.

5.2 **Packing.** The packing shall be level A, B, or C, as specified.

5.2.1 *Level A.*

5.2.1.1 *Generator set.* The generator set shall be packed by utilizing the steel base of the generator set as the base of the shipping container. A housing consisting of ends, sides, and top constructed in accordance with Specification MIL-C-104, type I, class optional, shall be provided to house the generator set. Clearance shall be provided at the ends, sides, and top, as specified in Specification MIL-C-104. The housing shall be secured to timbers bolted to the base of the generator set.

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5.2.1.2 *Repair parts and maintenance tools.* Unless otherwise specified, on-board repair parts and maintenance tools shall be packed for drydock fitted with bin or drawer stowage. When repair parts and maintenance tools are destined for a drydock that is fitted with bin or drawer type stowage, the repair parts and maintenance tools shall be packed in close-fitting boxes conforming to Specification PPP–B–601 (overseas type) , or PPP–B–621 (class 2). When repair parts and maintenance tools are destined for a drydock that is not fitted with bin or drawer type stowage, the repair parts and maintenance tools shall be packed in repair parts boxes conforming to Specification MIL–B–233, and overpacked in boxes specified above. Contents shall be cushioned, blocked, and braced to prevent movement within the boxes. The gross weight of each box shall not exceed approximately 200 pounds, except where the weight of an individual item exceeds this amount. Strapping shall be flat steel, type I, class B.

5.2.1.3 *Accessory equipment.* Accessory equipment not mounted and attached to the basic generator set, shall be packed in close-fitting boxes conforming to Specification PPP–B–621 (class 2) or PPP–B–601 (overseas type). The gross weight of each box shall not exceed approximately 200 pounds, except where the weight of an individual item exceeds this amount, but not over 500 pounds. Individual items that weigh over 500 pounds shall be packed in crates conforming to Specification MIL–C–104, type I, style a, class optional. Contents shall be cushioned, blocked, braced, and anchored to prevent movement within the container.

5.2.2 *Level B.* The generator set, repair parts, maintenance tools, and accessory equipment shall be packed as specified for level A, except that nailed wood and cleated-plywood boxes shall conform to Specification PPP–B–621 (class 1) or PPP–B–601 (domestic type), respectively. Strapping shall be either class A or B.

5.2.3 *Level C.* The complete generator set shall be packed or prepared for shipment in a manner to insure arrival at destination in satisfactory condition and which will be acceptable to the carriers at lowest rates.

5.3 **Marking.** Interior packages and shipping containers shall be marked in accordance with Standard MIL–STD–129.

6. NOTES

6.1 **Intended use.** The generator sets covered by this specification are intended to be used with floating drydocks.

6.2 **Ordering data.** Procurement documents should specify the following:

(a) Title, number, and date of this specification.

(b) Kilowatt rating required (see 1.2).

(c) Data required to compute formulae in 3.7.

(d) Whether sea water pump is to be furnished (see 3.8.1).

(e) Type of cranking system required (see 3.8.5).

(f) Number of compressors required when compressed air starting is specified (see 3.8.5.2.1).

(g) Technical manuals required if other than specified (see 3.17).

(h) Length of time inspection records shall be kept available to the Government (see 4.1).

(i) Whether paralleling test with identical generator at place of manufacture is acceptable (see 4. 6.3 (a)).

(j) Level of preservation and packaging and level of packing required (see 5.1 and 5.2).

(k) When drydocks, for which the generator sets are destined are not fitted with bin or drawer type stowage (see 5.2.1.2).

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